Step 1: Software Development Projects

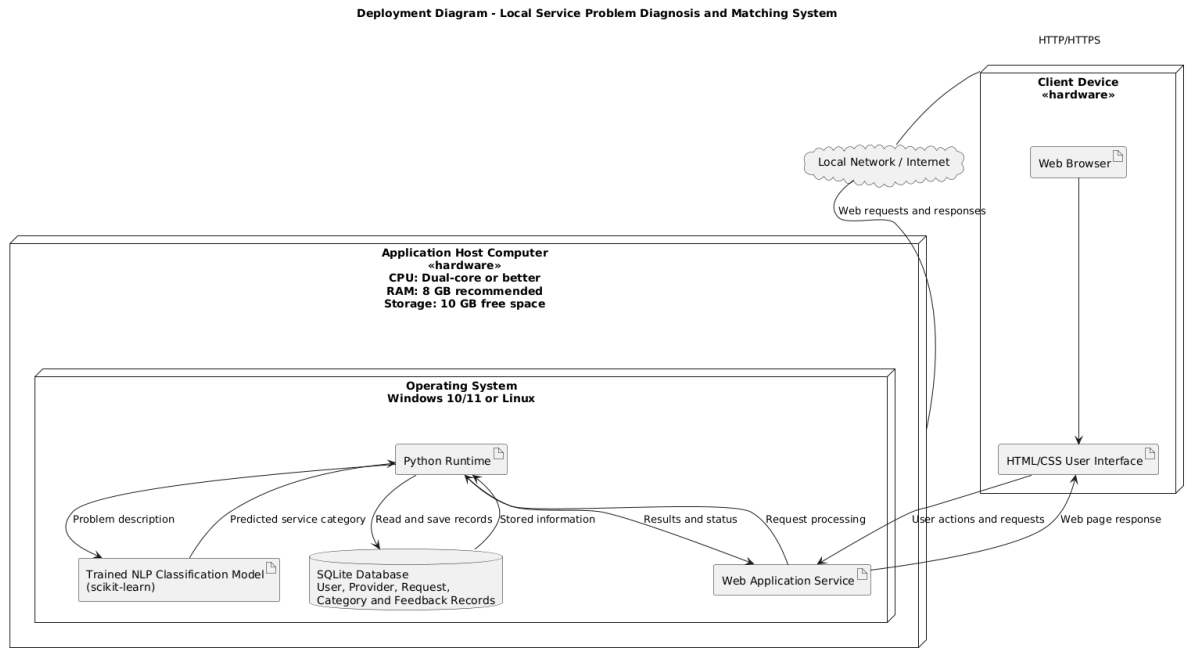
Project Title:
Local Service Problem Diagnosis and Matching System

5. System Architecture

The system uses a three-layer structure consisting of the presentation layer, application layer, and data layer. The presentation layer provides HTML and CSS pages through which local users, service providers, and administrators access the application. The application layer uses Flask to handle login, service requests, provider matching, and request tracking.

The NLP component uses a trained scikit-learn model to predict a suitable service category from the user's problem description. The provider-matching component uses that category to find registered providers offering the required service. Users can also choose a category directly when they already know which service they need.

The data layer uses SQLite to store account details, provider information, service requests, categories, request statuses, and feedback. Flask components access this database when information needs to be saved or retrieved. The administrator manages accounts, service categories, and request records through protected pages.

The system provides category suggestions to help users find suitable providers. It does not guarantee an exact technical diagnosis of the reported problem.

Deployment Diagram